Bilkent University EE313 ECD Lab 3 Work Report

Single-Supply Push-pull Class-B Power Amplifier

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Introduction and Aim

In this lab experiment, our aim was to design a single-supply push pull Class B power amplifier with given specifications. The gain must be 20+mod([ID] ,7)dB. My gain is 23dB.

To chose Vin, I looked at the required output voltage which is 16Vpp. To get a gain of 23dB, I chose my Vin as 0.57V. This way $0.57 \times 14.12 = 8.04V$ would be achieved. This would create a 16.08V Vpp which is almost equal to our required value.

Specifications and Results

Firstly, here is the whole circuit design:

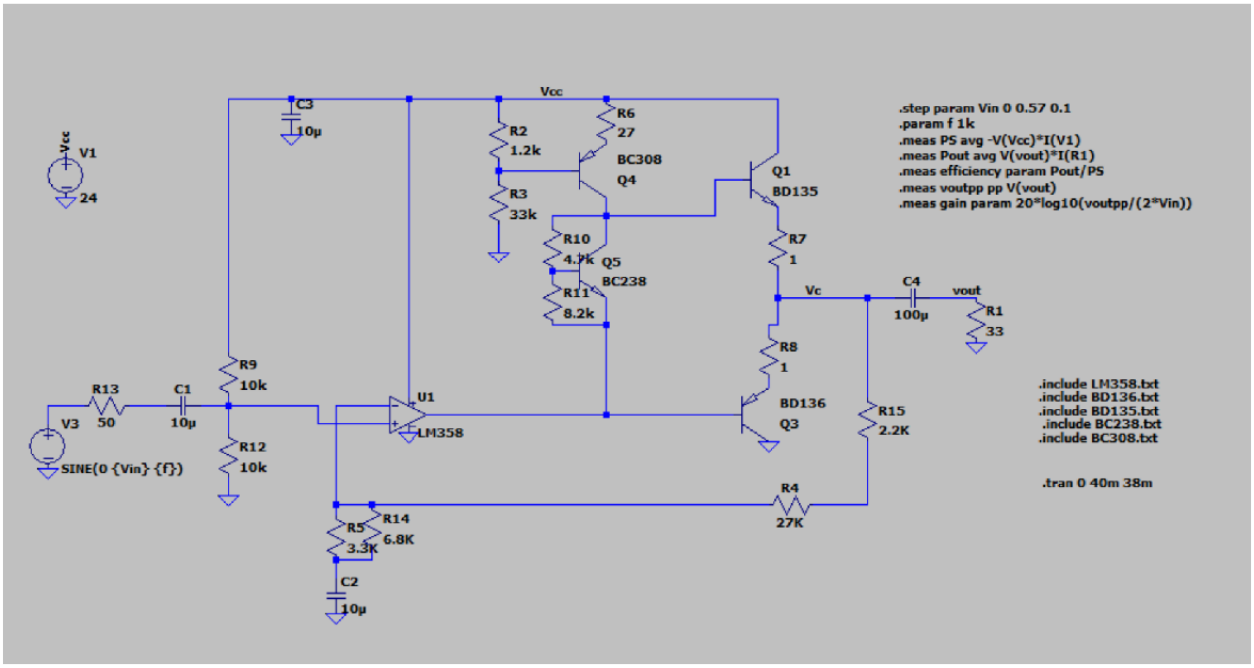


Figure 1: Circuit Design on LTSpice

To satisfy all of the specifications, I chose my resistor values wisely. I chose R4 and R5 resistors carefully to get the desired gain. R4 is 2 resistors valued at 27KΩ and 2.2KΩ which is in total 29.2KΩ. R5 is chosen ad 2 resistors in parallel which are 3.3KΩ and 6.8KΩ which gives in total around 2.22KΩ of resistance. Our gain is the following

$$1 + \frac{R_4}{R_5} = 14.12$$

So the values chosen give us around 14.15 gain which corresponds to a gain of 23.01dB which is almost equal to 23dB.

R2 and R3 values are found 1.2KΩ and 33KΩ respectively. To get the required V_{CE} value for a Class B Amplifier, R10 is chosen as 4.7KΩ and R11 is chosen as 8.2KΩ.

A large capacitor with 100µF of capacitance is used on the output to be used as an large-value DC Block Capacitor.

Now, here are the specifications and their explanations:

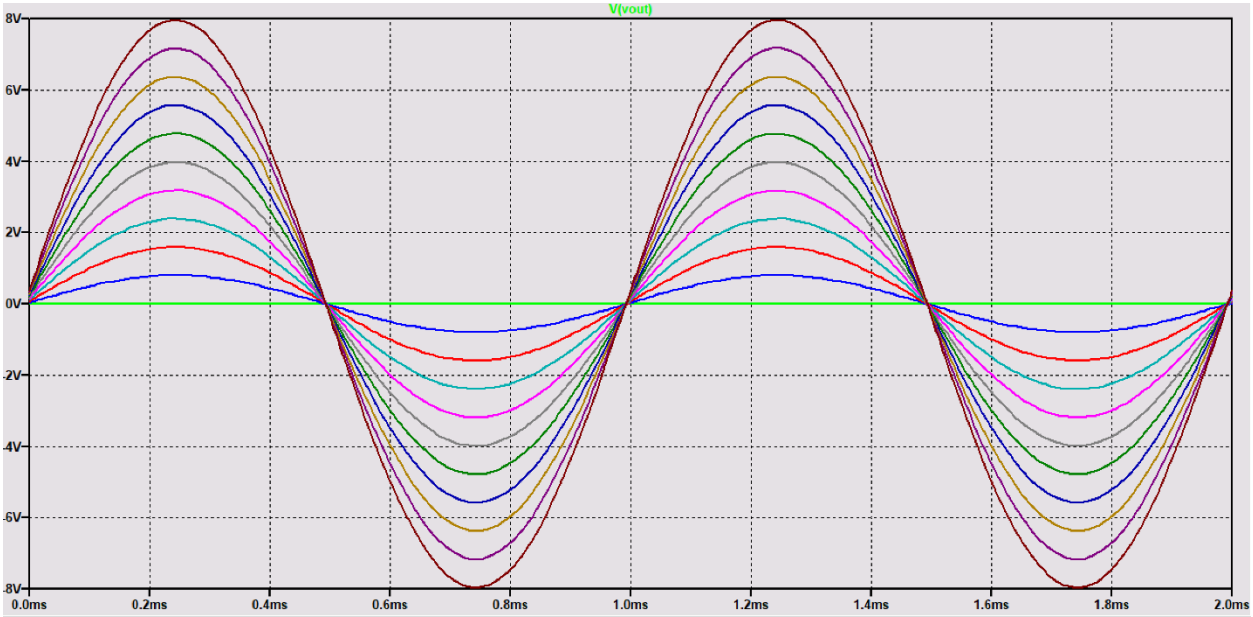


Figure 2: Plot of Vout when Vin is varying 0V and 0.57V with 0.057V Steps

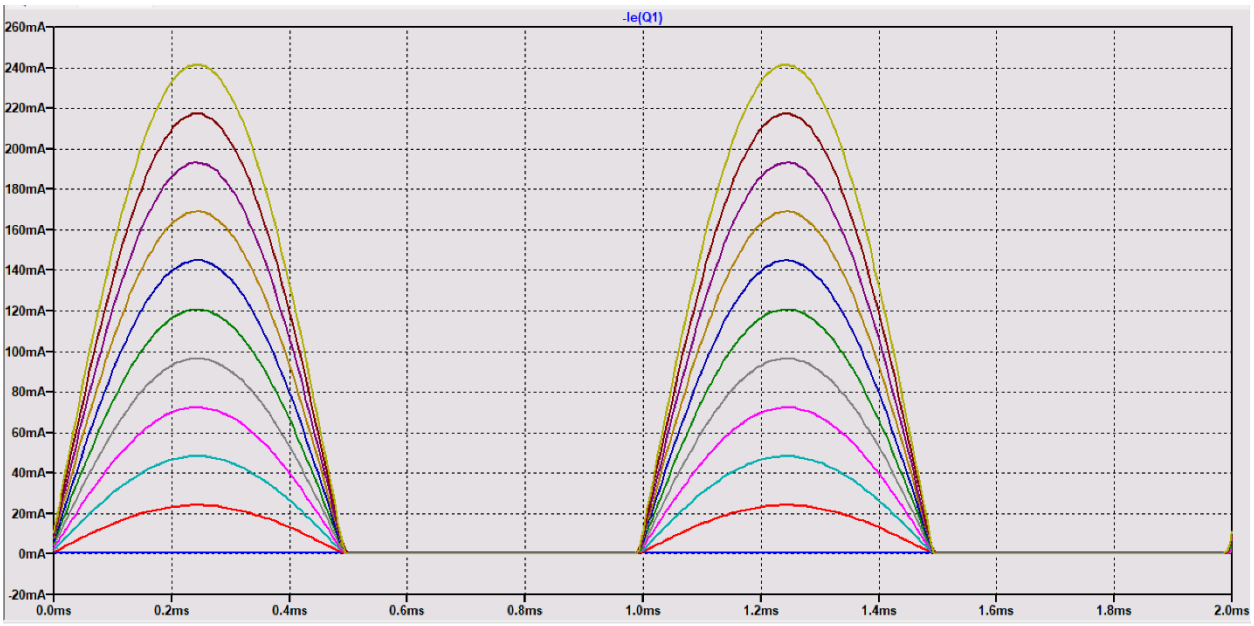


Figure 3: Plot of -Ie(Q1)

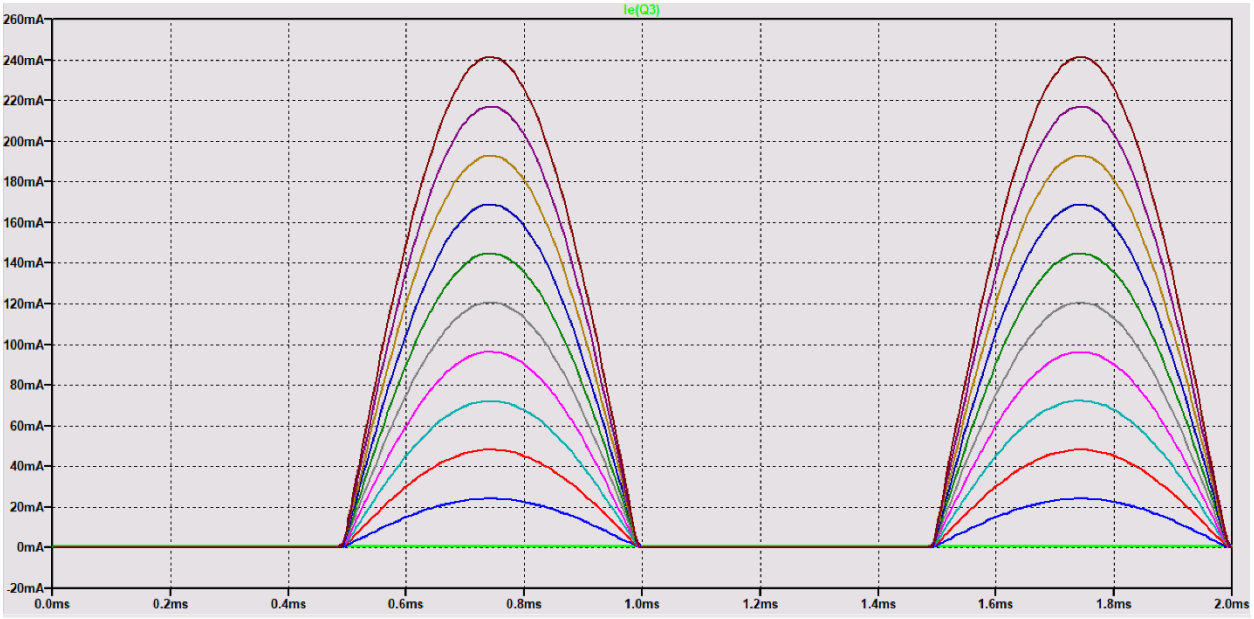


Figure 4: Plot of Ie(Q3)

1. The amplifier should deliver at least 0.95W power to a 33Ω resistance (16Vpp to a 33Ω power resistor) at 1KHz with the chosen gain value.

Measurement: voutpp

step	PP(v(vout))	FROM	TO
1	4.13738e-10	0	0.002
2	1.59292	0	0.002
3	3.18645	0	0.002
4	4.77973	0	0.002
5	6.37287	0	0.002
6	7.96524	0	0.002
7	9.55804	0	0.002
8	11.1519	0	0.002
9	12.7435	0	0.002
10	14.3333	0	0.002
11	15.9332	0	0.002

Figure 5: Vout with each step.

Measurement: pout

step	AVG(v(vout)*i(r1))	FROM	TO
1	1.75799e-21	0	0.002
2	0.00961156	0	0.002
3	0.0384466	0	0.002
4	0.0864672	0	0.002
5	0.153676	0	0.002
6	0.240055	0	0.002
7	0.345653	0	0.002
8	0.470443	0	0.002
9	0.614399	0	0.002
10	0.777628	0	0.002
11	0.960342	0	0.002

Figure 6: Pout with each step.

It can be seen that from these 2 Figures, the output power is around 0.9603W at the last step which has 15.9332Vpp which is almost equal to the required Vpp, 16V. The output power is bigger than 0.95W which satisfies the specification.

2. The harmonics (the highest is possibly the third harmonic) at the 0.95W output power level should be at least 40 dB lower than the fundamental signal at 1 KHz.

When at Vin is 0.57V which is at the 0.96W output power level:

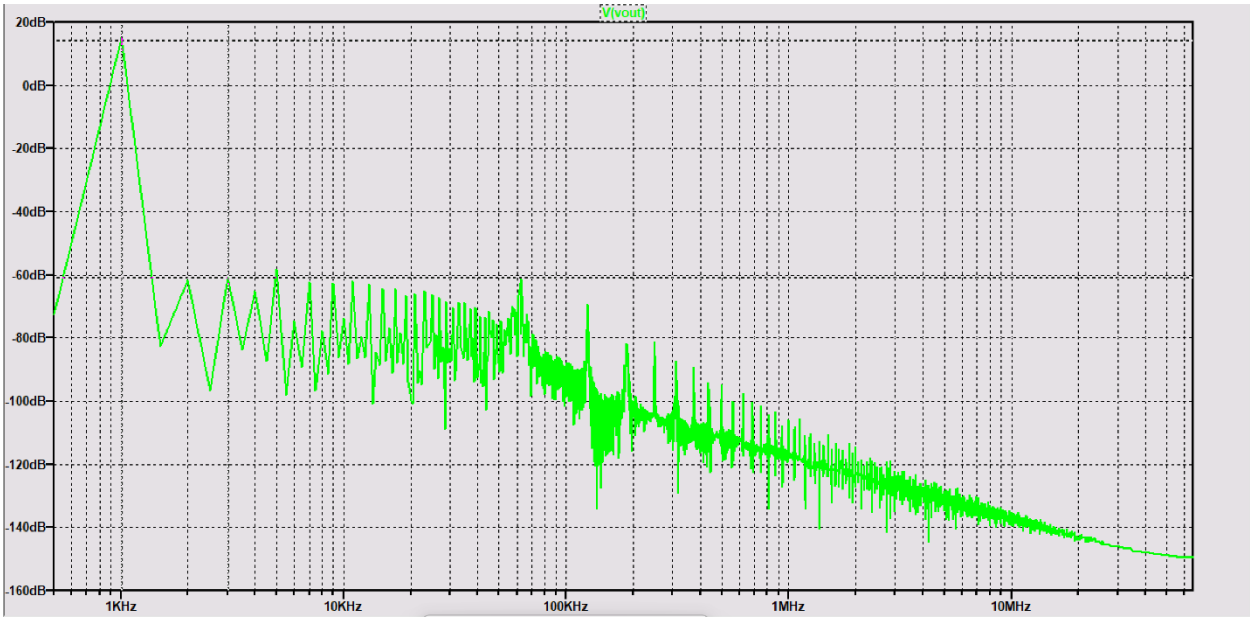


Figure 7: Difference of magnitude between fundamental and the third harmonic.

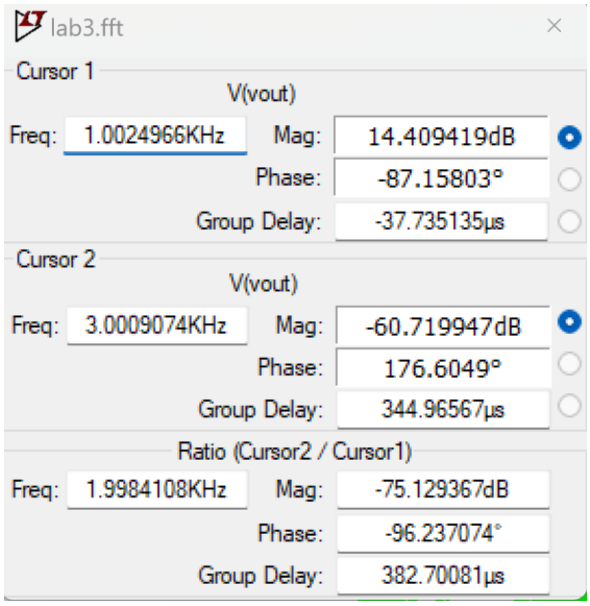


Figure 8: -75.129dB difference between the the fundamental(1KHz) and the third harmonic(3KHz).

As it can be seen from the plot, the difference is around 75dB which is more than a 40dB difference so this satisfies our specification.

3. The power consumption at quiescent conditions should be less than 500mW.

To satisfy this criteria, we can take a look at the first step, where it is the zero input condition. By looking at ps on the given data:

Measurement: ps

step	AVG(-v(vcc)*i(v1))	FROM	TO
1	0.212316	0	0.002
2	0.380987	0	0.002
3	0.5638	0	0.002
4	0.746701	0	0.002
5	0.929666	0	0.002
6	1.11246	0	0.002
7	1.2954	0	0.002
8	1.47819	0	0.002
9	1.66041	0	0.002
10	1.84388	0	0.002
11	2.02639	0	0.002

Figure 9: ps measurement data.

We can clearly see that the ps value at the first step is 0.2123W which is smaller than 0.5W or 500mW. This value clearly satisfies our third specification.

4. The amplifier's efficiency (output power/total supply power) should be at least 40% at max power output (0.95W) at 1KHz.

To check this specification we can take a look at the efficiency measurement data given below:

Measurement: efficiency

step	pout/ps
1	8.28007e-21
2	0.0252281
3	0.0681919
4	0.115799
5	0.165302
6	0.215788
7	0.26683
8	0.318255
9	0.370029
10	0.421734
11	0.473918

Figure 10: Efficiency measurement data.

As it can be seen, at the last step(max power output), our efficiency is around 0.47 which is bigger than 0.4 or 40% so we can say that this specification is also satisfied.

5. The −3dB bandwidth of the amplifier should be at least 150Hz to 15KHz.

To check this, we use a frequency function for the maximum input level(0.57V). To do this, I used .step param f 150 15k 1485 function to see the change between 150Hz and 15kHz with equal steps of 1485Hz. Then, we will take a look at the gain measurement data. They must be greater than 20dB because of the -3dB bandwidth.

Measurement: gain

step	20*log10(voutpp/(2*vin))
1	20.221
2	22.9043
3	22.894
4	22.8852
5	22.887
6	22.8915
7	22.8909
8	22.8677
9	22.8626
10	22.7961
11	22.5941

Figure 11: Gain with different frequency steps between 150Hz to 15kHz

We can see that with every step, the gain is bigger than 20dB so our last specification is also satisfied.

Conclusion of the first part:

In this part, we designed and analyzed a class B power amplifier with a gain of 23dB and a bandwidth of 150-15kHz. Overall, we simulated the circuit on LTSpice and tried to satisfy the given specifications. In the end, all of the given criteria were satisfied and the circuit operated as expected. Now we can say that it is ready to be implemented on hardware on the second part of the lab experiment.

Experimental Work

Now we implemented this circuit on hardware and tried to get the same specifications on the circuit that we built. Here is the circuit that is implemented on a breadboard:

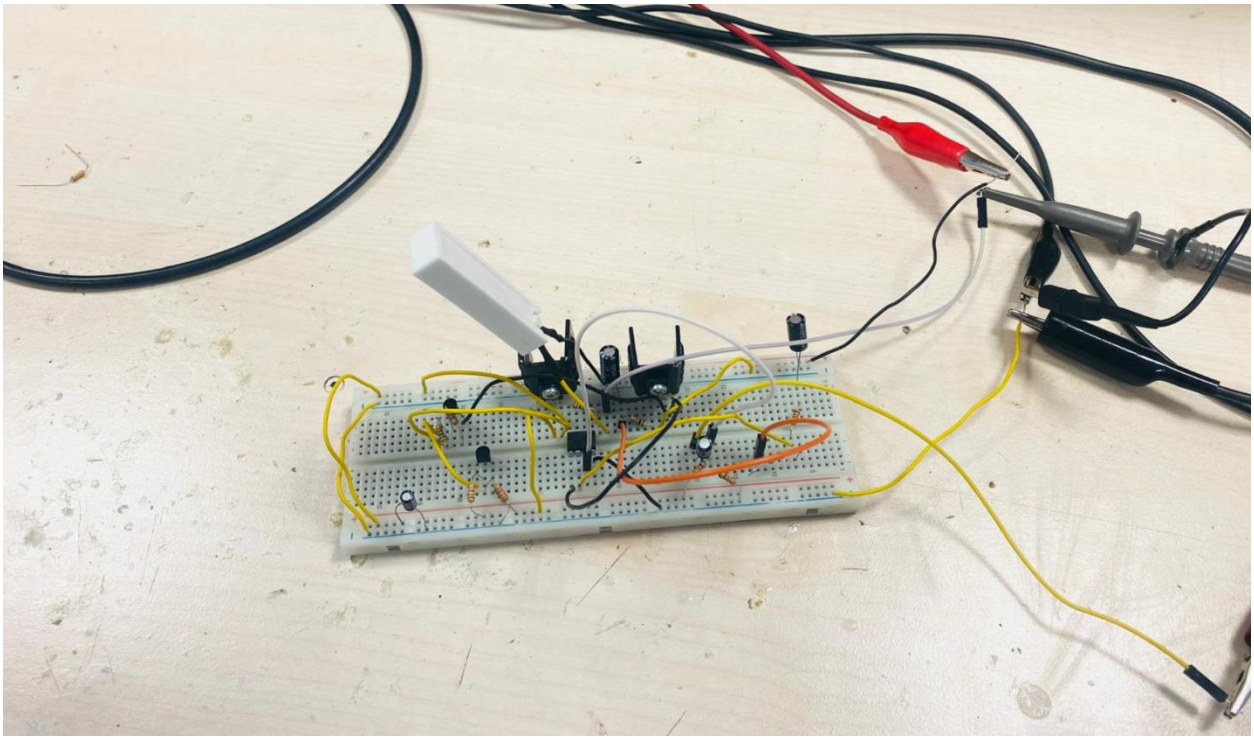


Figure 12: Implemented design on hardware.

After implementing, it is now time to test this circuit. I set Vin as 0.565V as calculated, set the frequency as 1kHz on the signal generator and connected my oscilloscope probe to the output to see the output voltage plot.

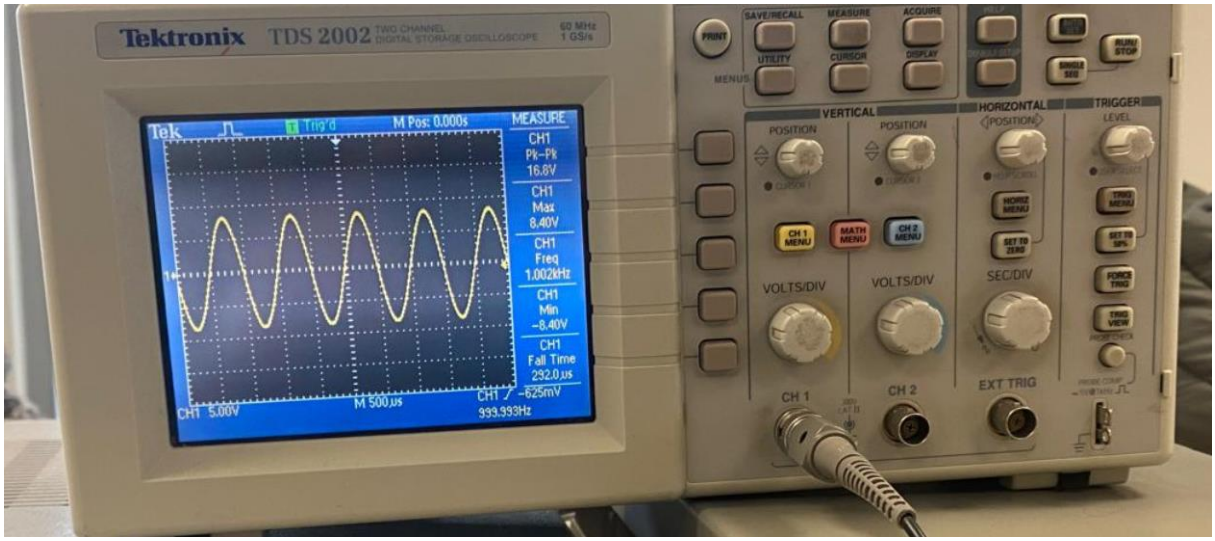


Figure 13: Vout at 1kHz frequency.

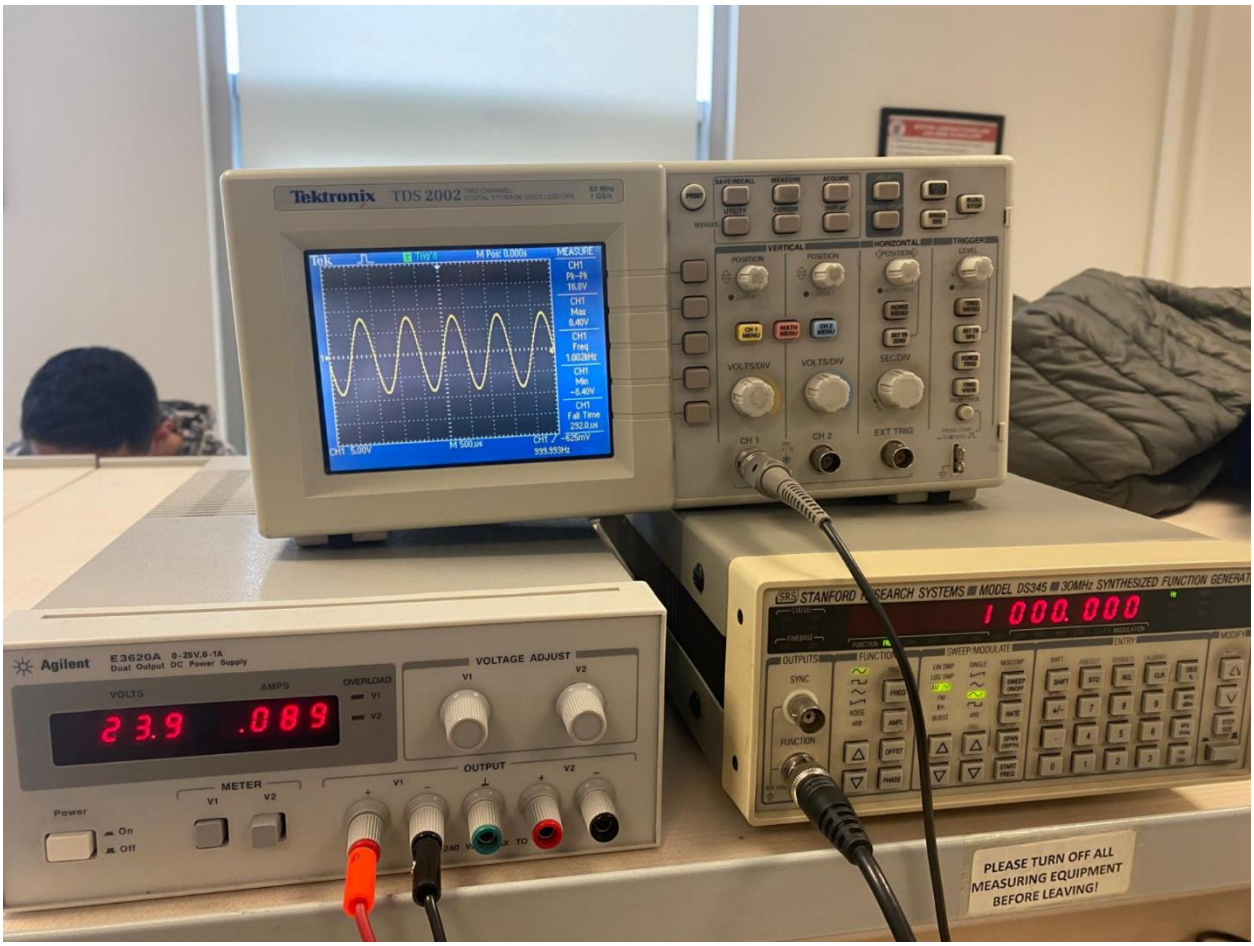


Figure 14: Vout at 1kHz frequency.

As it can be seen from the plot Vout is 16.8Vpp. We can calculate the power delivered by using the formula below:

$$P = \frac{(16.8)^2}{8 * 33} = 1.069W > 0.95W$$

Thus, the first requirement is satisfied.

To show the second requirement, we can use the math menu of the oscillator. Here below, we can see the harmonics:

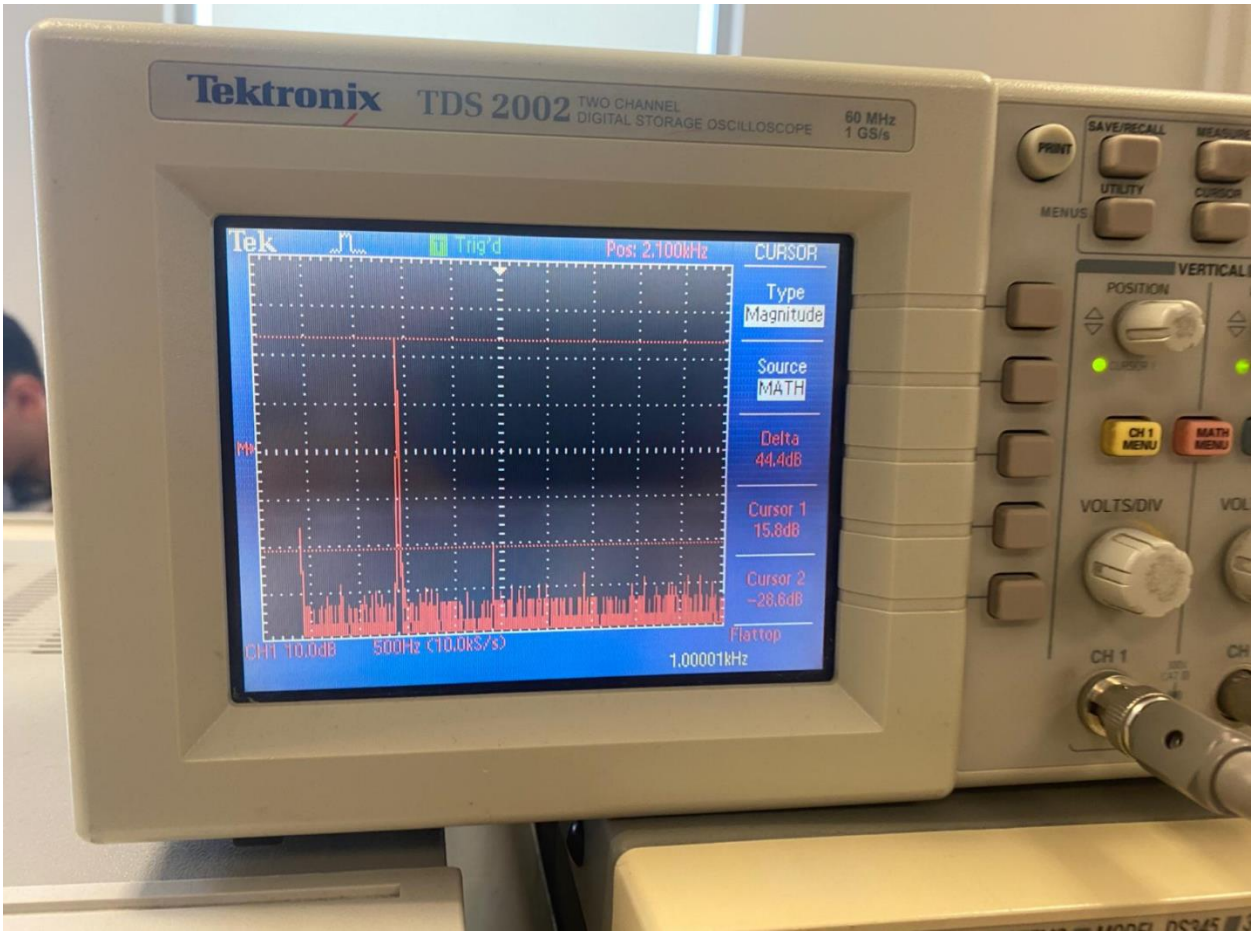


Figure 15: The harmonics.

We can see that the difference between the harmonics is 44.4dB>40dB, thus the second requirement is also satisfied.

Now, for the third requirement, we calculate the power consumption at Vin=0, quiescent conditions.

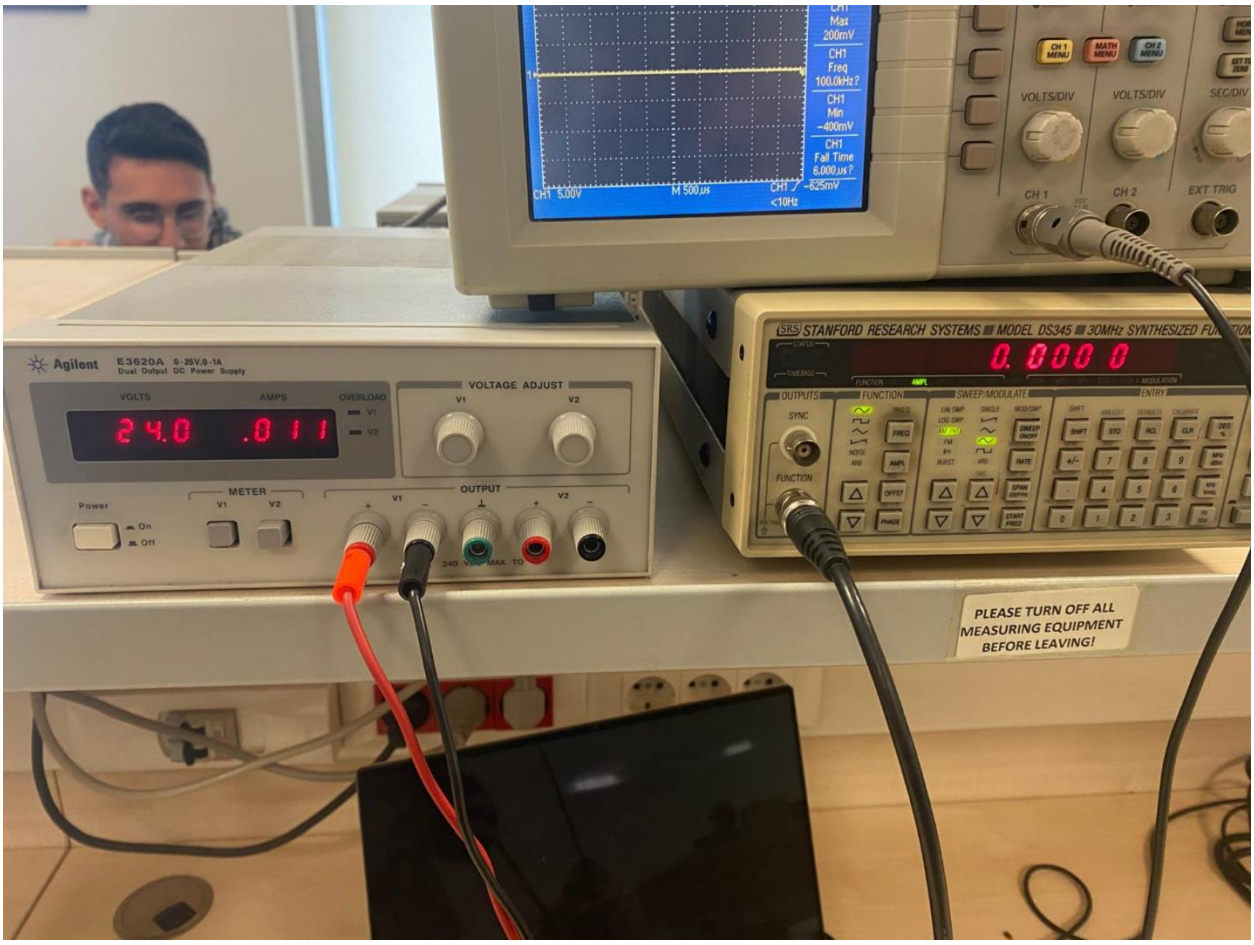


Figure 16: Current supplied at quiescent conditions.

As it can be seen from the figure, the current is 11mA. So we can calculate the power by:

$$P = 24 * 0.011 = 0.264W < 0.5W$$

Thus, third condition is also satisfied.

To calculate the efficiency we calculate, P_D/P_S. We already calculated the power delivered. We can see and calculate the power delivered on Figure 1. $P_S = 24 * 0.089 = 2.136W$. Therefore the efficiency can be calculated by:

$$Efficiency = \frac{1.069}{2.136} = 0.5004 = 50.04\% > 40\%$$

Therefore, this fourth requirement is also satisfied.

For the last requirement:

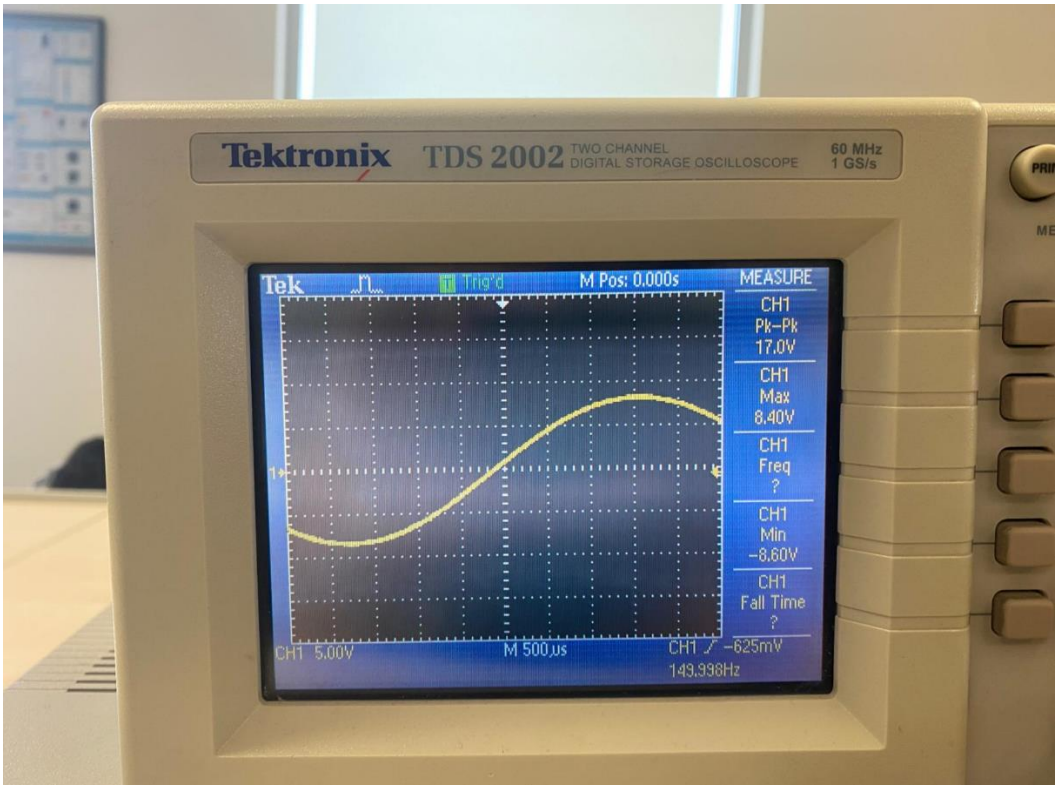


Figure 17: Vout at 150Hz.

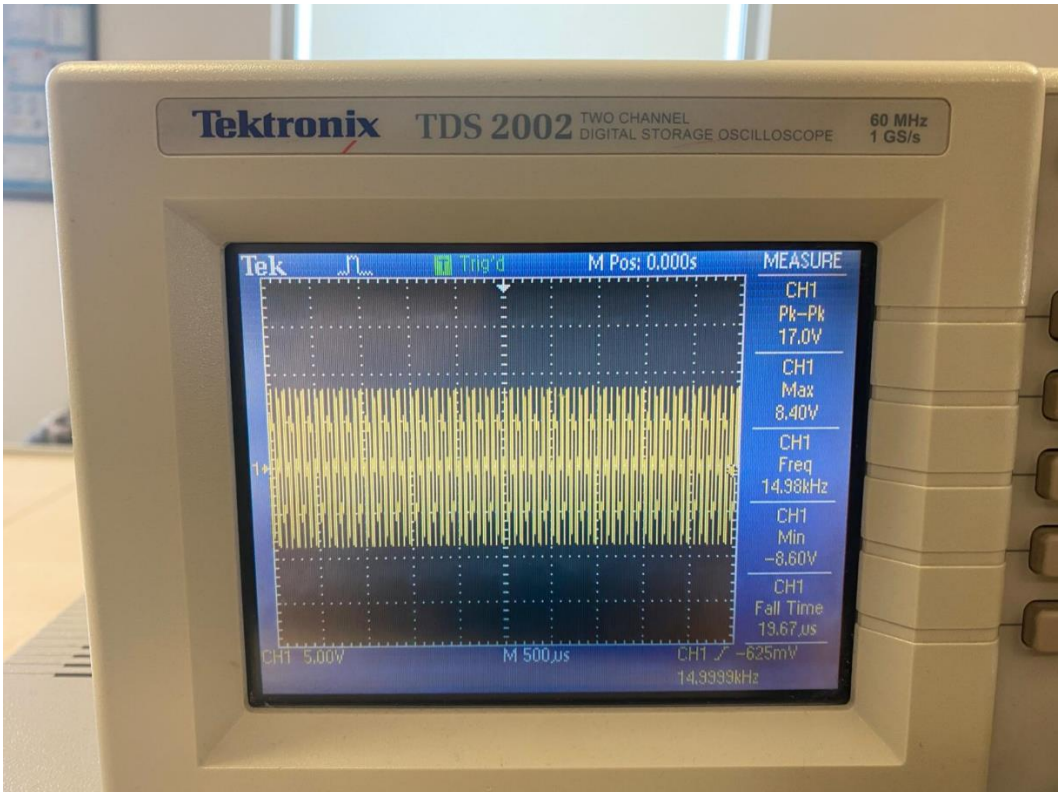


Figure 18: Vout at 15kHz

By these plots we can calculate the gains for each frequency by:

$$20\log_{10}\frac{8.5}{0.565} = 23.54dB\ for\ f = 150Hz$$

$$20\log_{10}\frac{8.5}{0.565} = 23.54dB\ for\ f = 15kHz$$

Therefore, both are in the range -3dB bandwidth. The last requirement is now satisfied. Here below, there are different frequencies to plot the gain as a function of frequency at max input level.

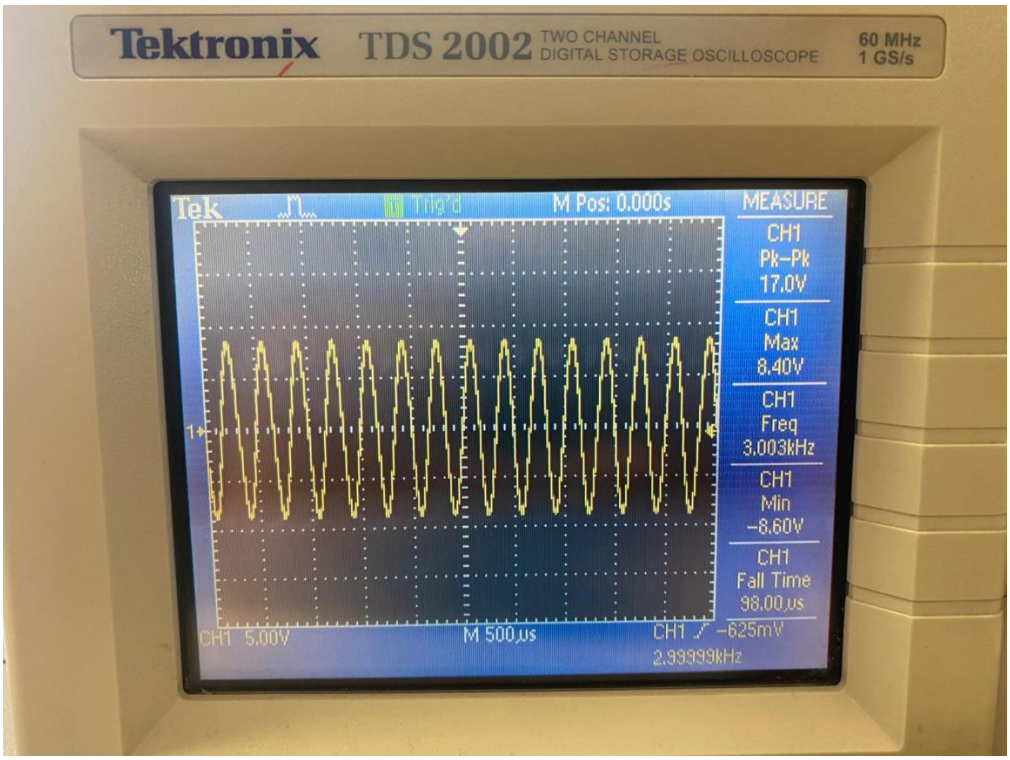


Figure 19: Vout at f=3kHz

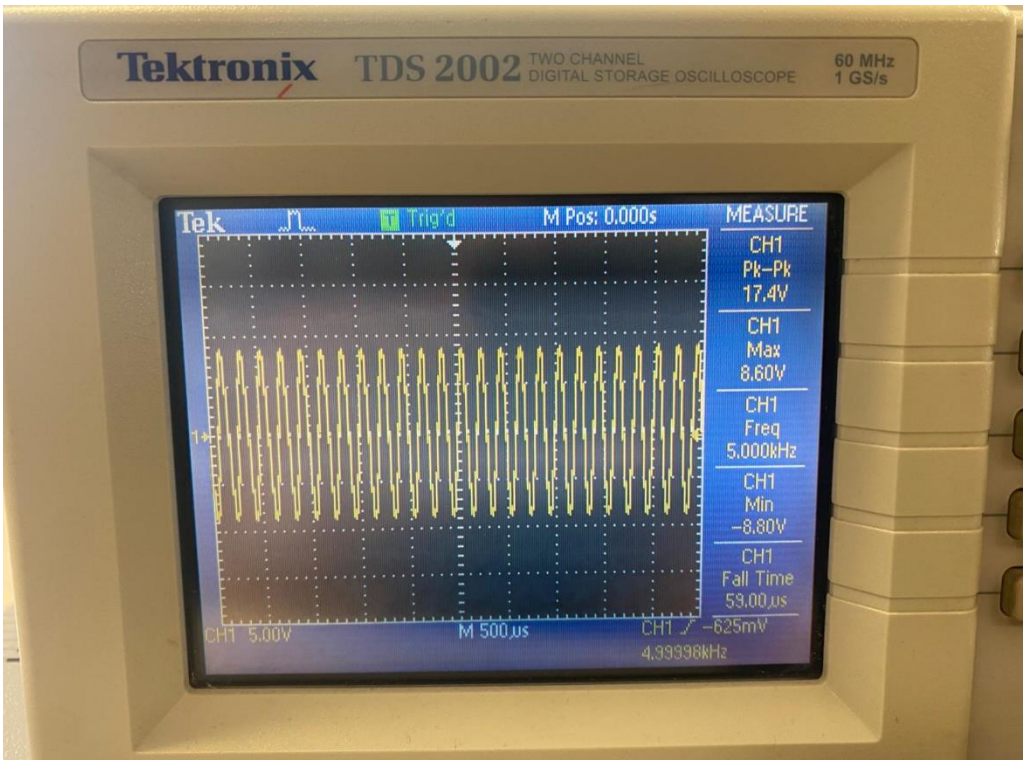


Figure 20: Vout at f=5kHz

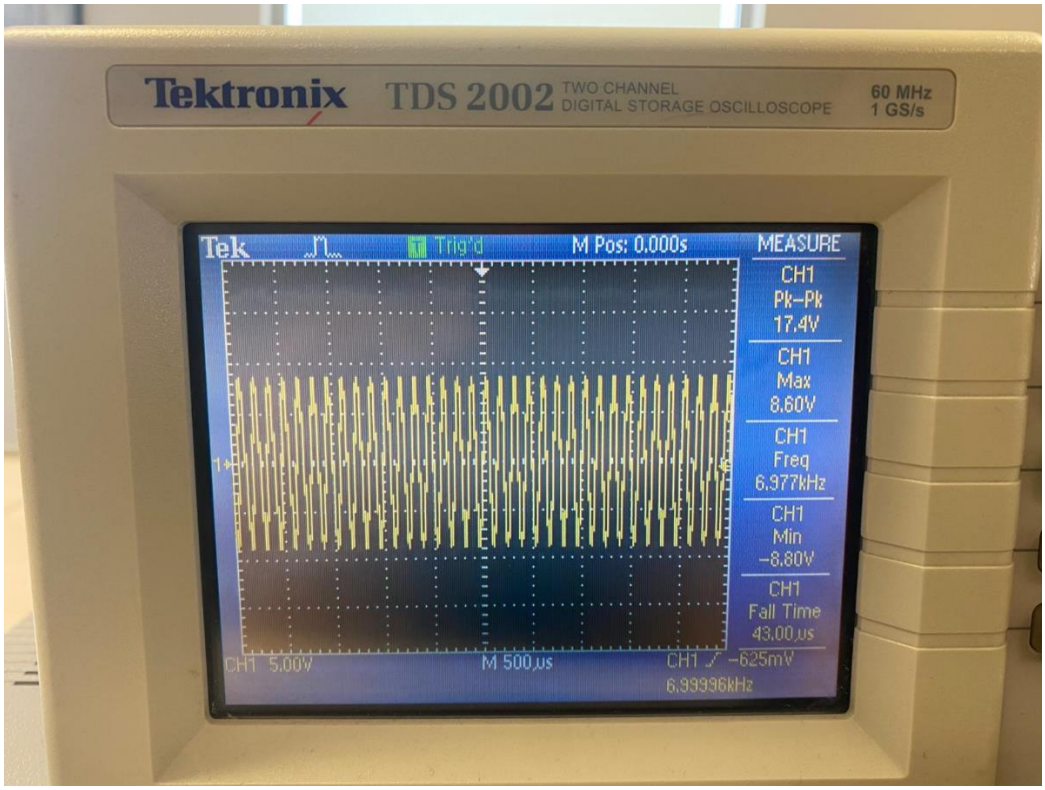


Figure 21: Vout at f=7kHz

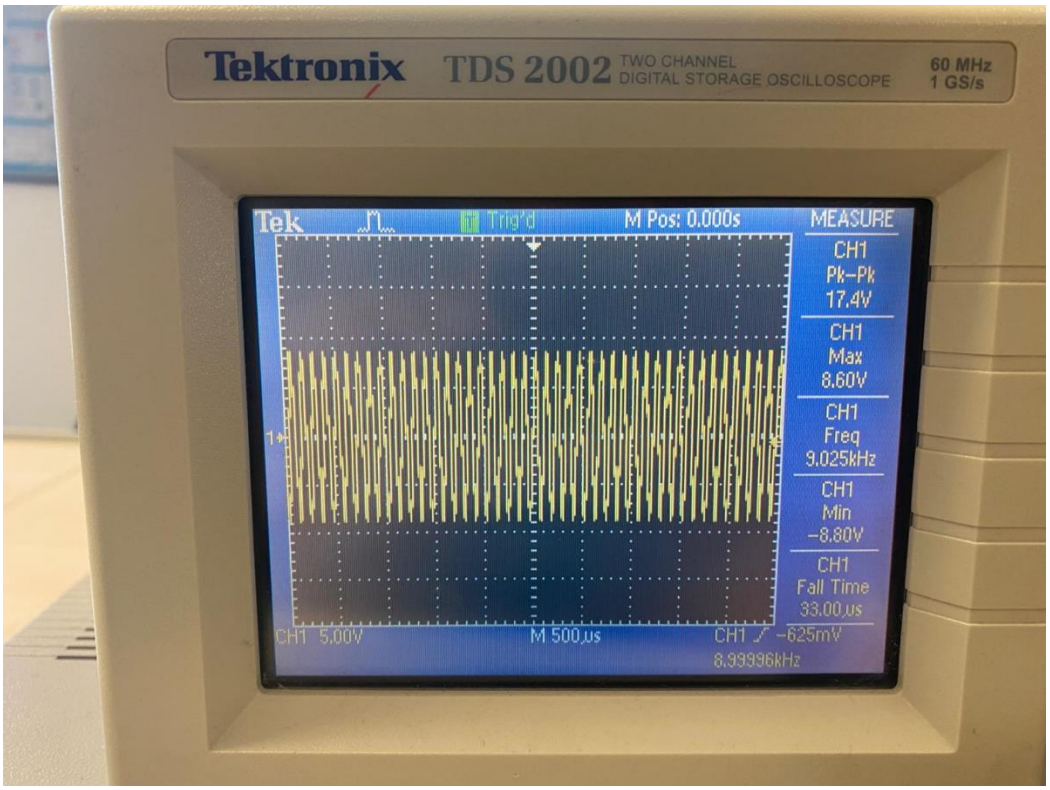


Figure 22: Vout at f=9kHz

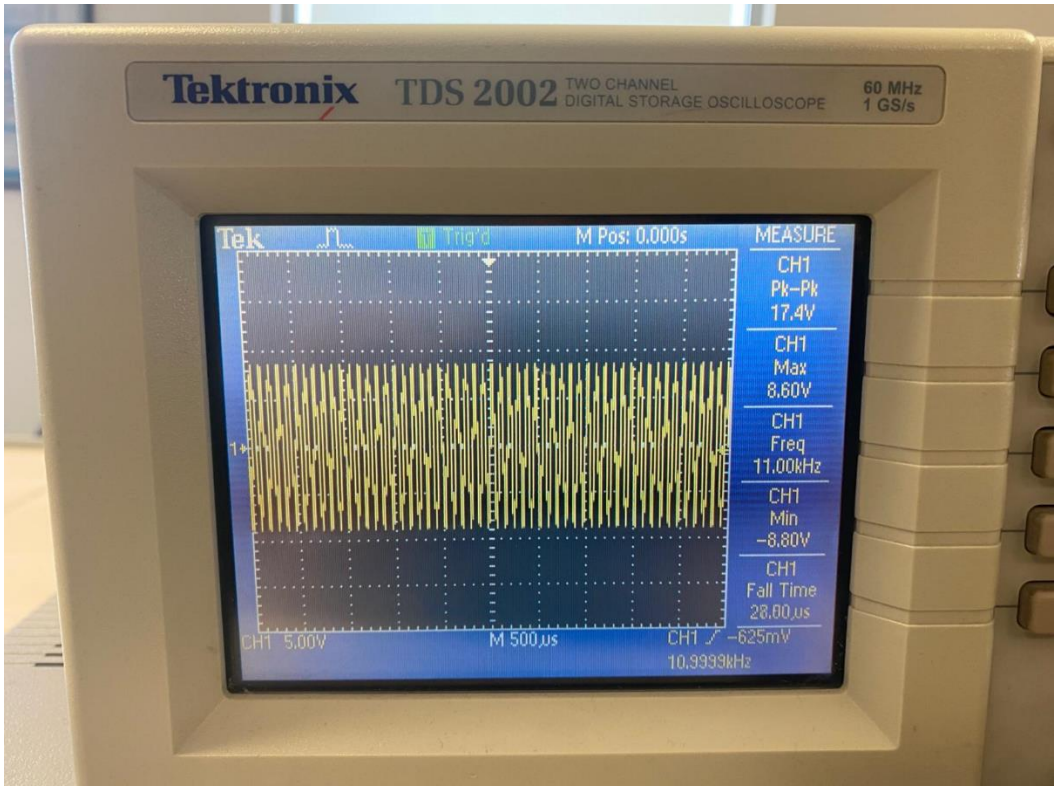


Figure 23: Vout at f=11kHz

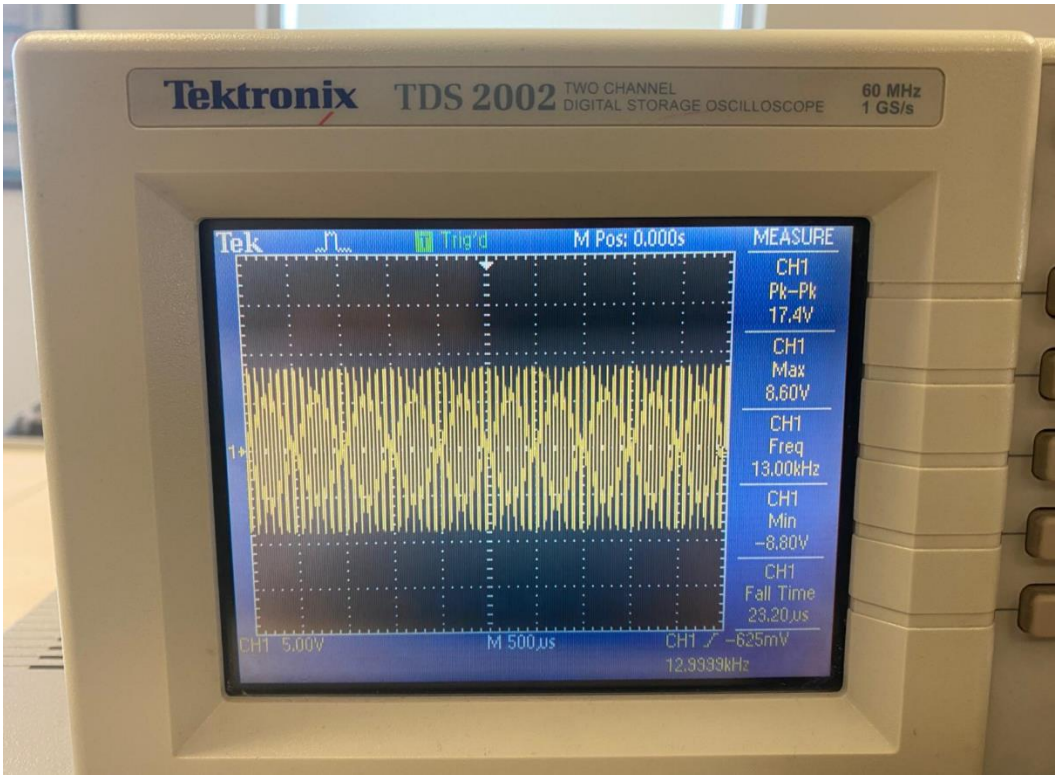


Figure 24: Vout at f=13kHz

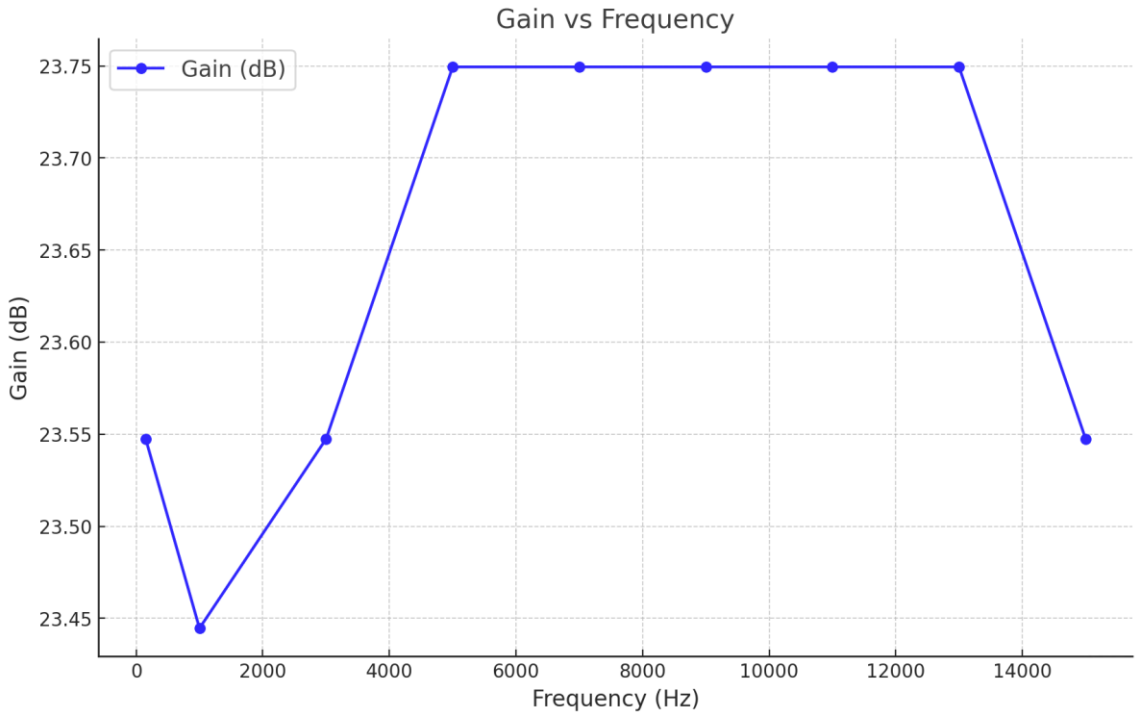


Figure 25: Gain plot as a function of frequency at max input level.

Additionally, to plot the efficiency for different input:

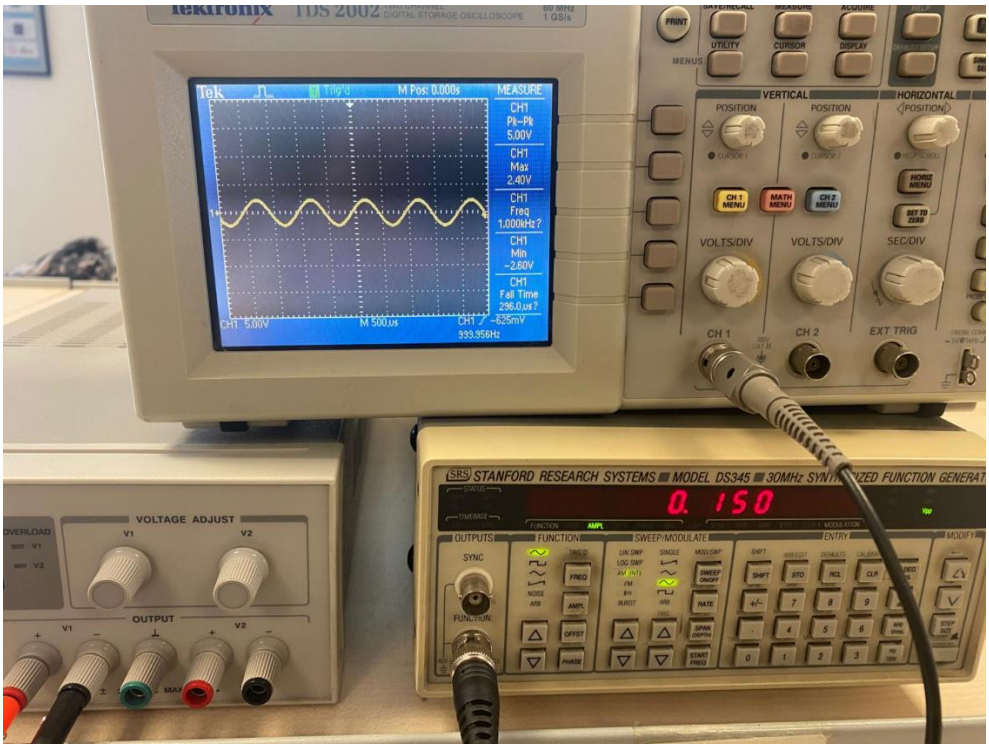


Figure 26: Vout at Vin=0.150Vpp

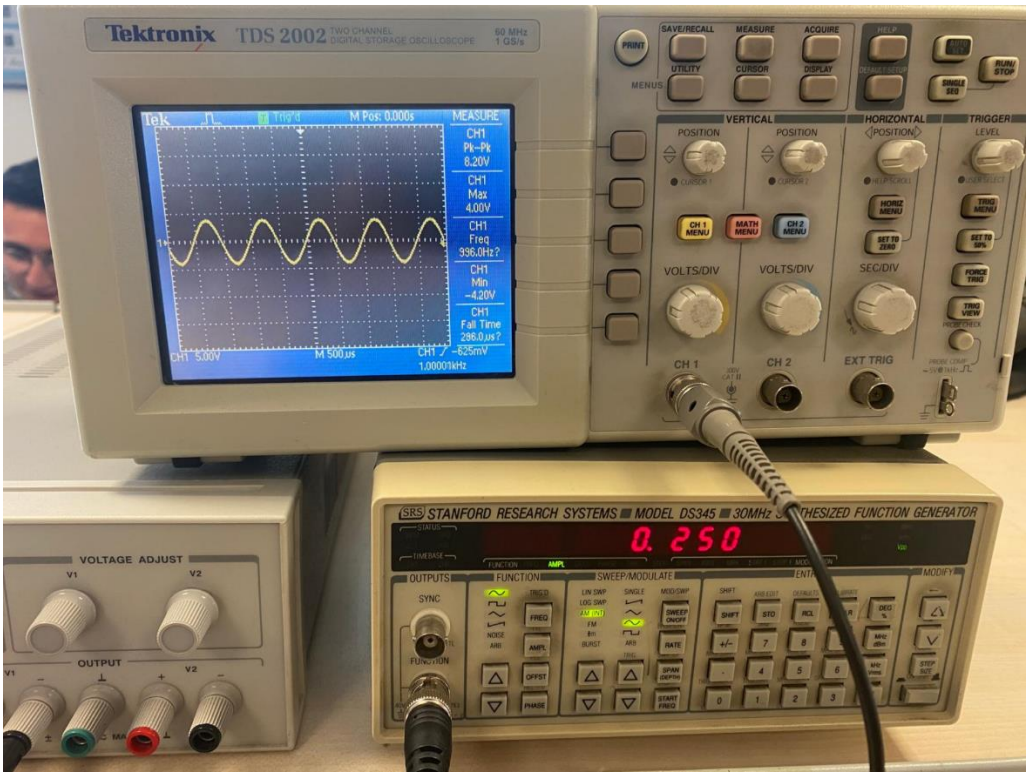


Figure 27: Vout at Vin=0.250Vpp

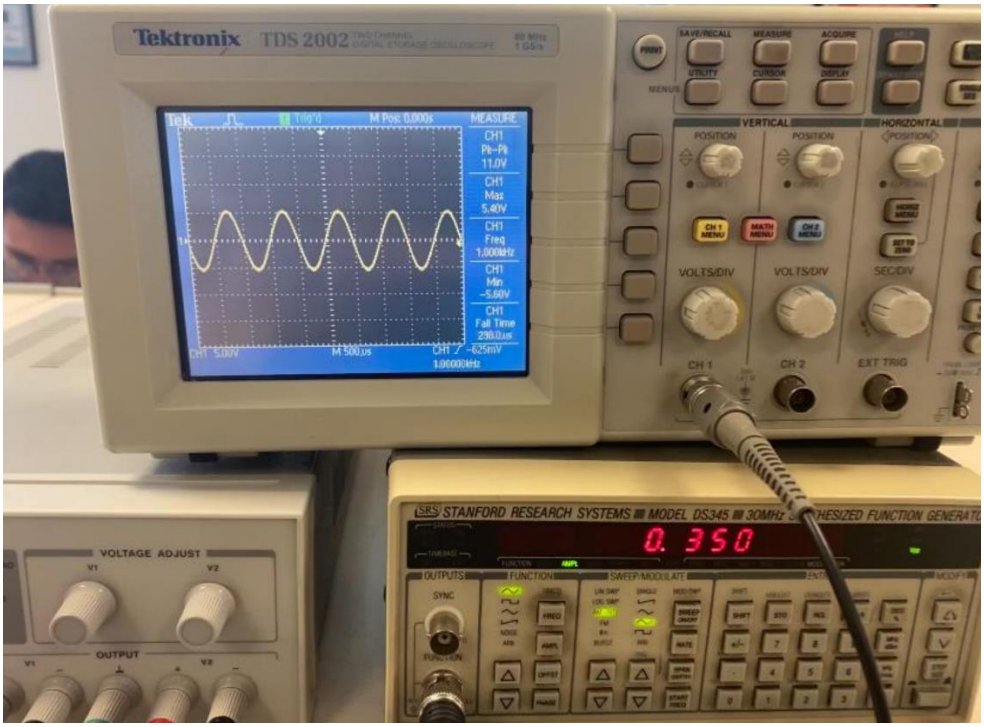


Figure 28: Vout at Vin=0.350Vpp

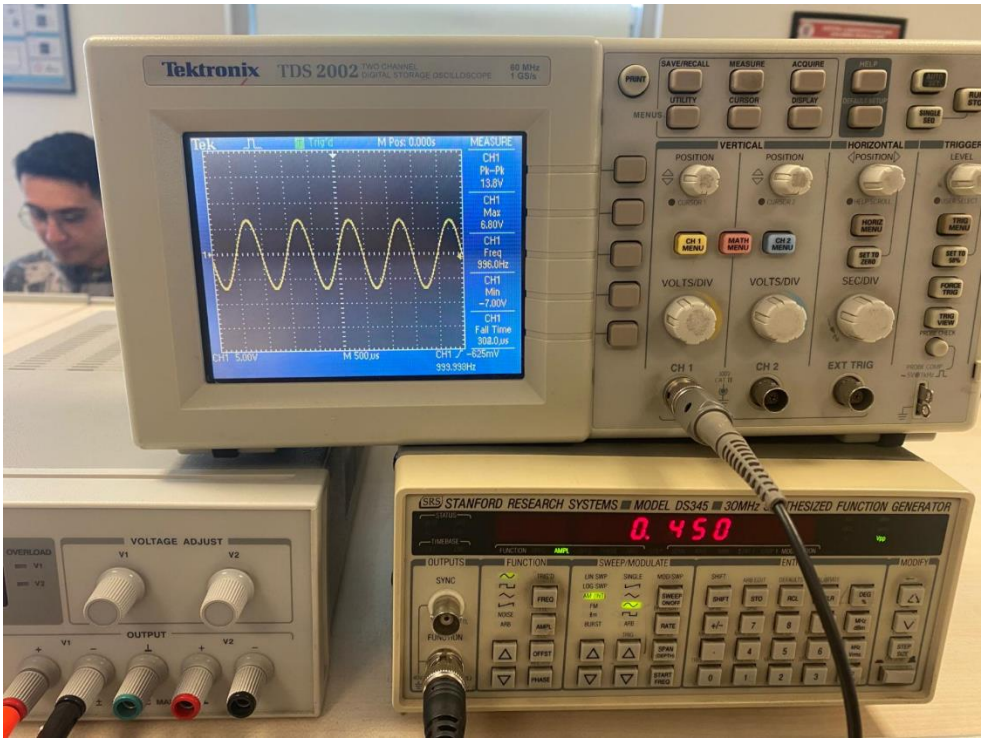


Figure 29: Vout at Vin=0.450Vpp

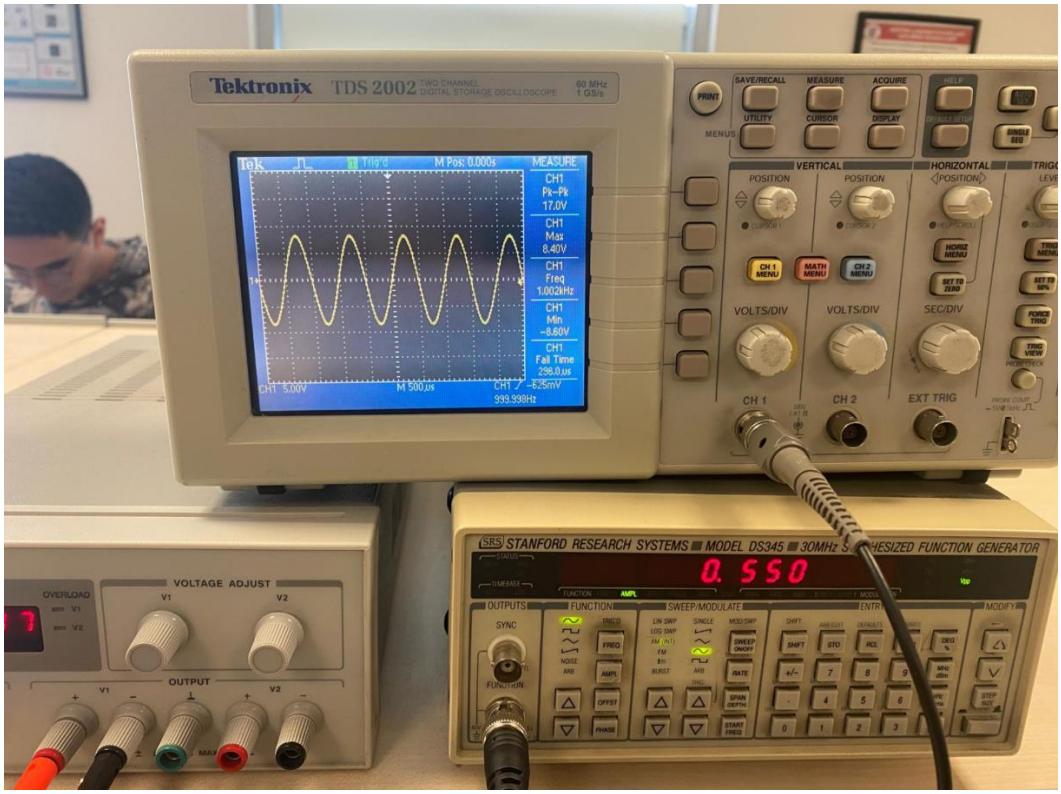


Figure 30: Vout at Vin=0.550Vpp

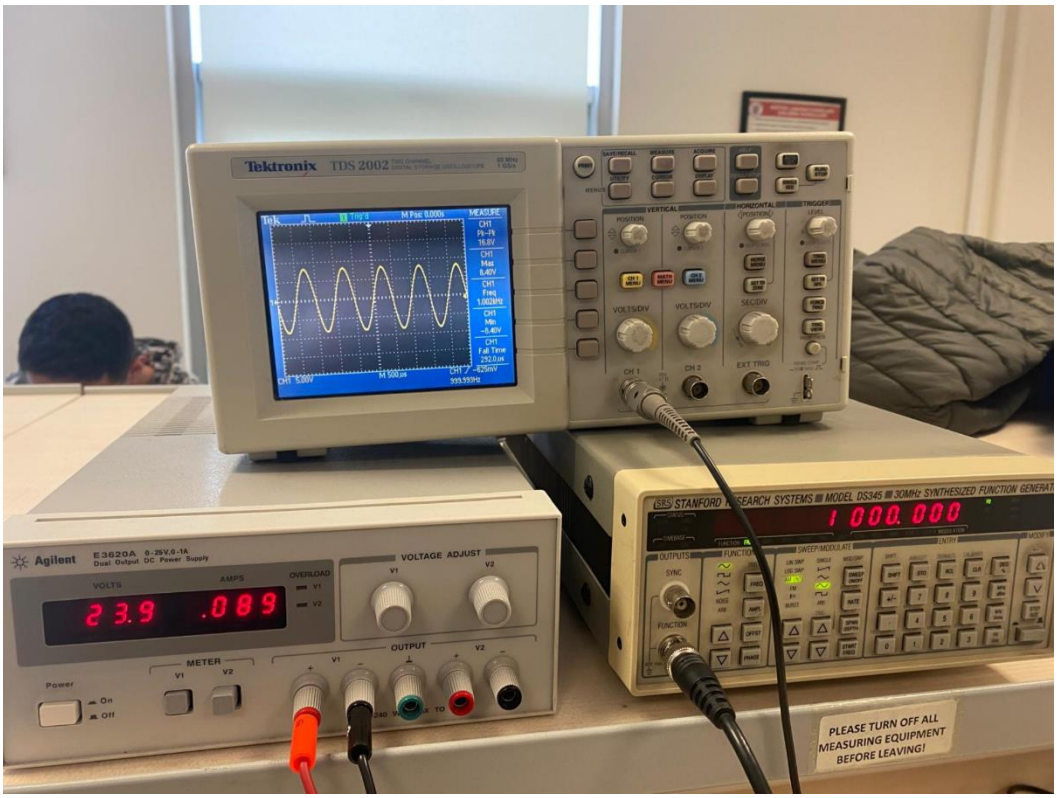


Figure 31: Vout at Vin=0.565Vpp

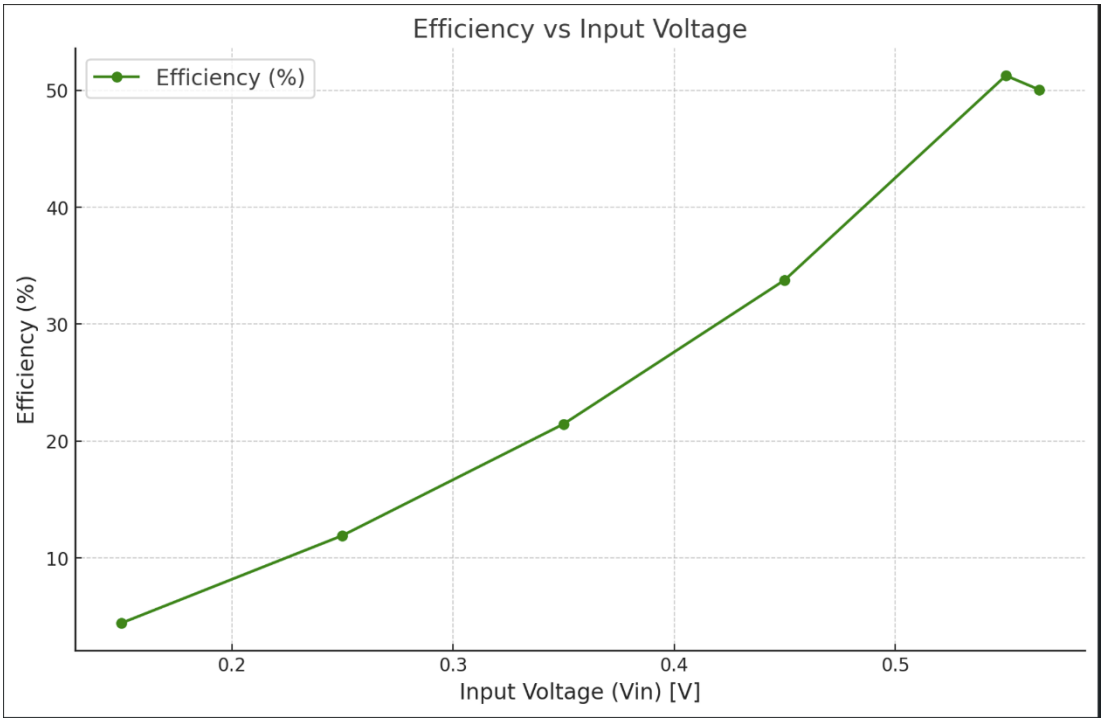


Figure 32:Efficiency plot as a function of input level.

Conclusion:

To conclude, all of the 5 requirements were satisfied. It was difficult to implement this design on hardware since it is very easy to connect the components in the wrong places because of the number of different nodes and components. However, the circuit worked properly with small errors. These can be because of the components, the cables used, or the human error. Overall, this experiment was completed successfully.